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Using device: mps
Using single-amplitude feyn data from /var/folders/1s/69py7bdx0g7g0t4_8rjh6g9h0000gn/T/single_amplit
ude_eval_823uj_o_/first_amplitude.csv
Imported 1 token rows
Wrote raw expression data to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML
/data_testing/outputs/ym_5pt_colour_500k_beam20.csv
Wrote normalized token data to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-
ML/data_testing/outputs/ym_5pt_colour_500k_beam20_tok.csv
Input token limit: unlimited; longest input: 2151 content / 2153 with BOS/EOS
Numerical evaluator: ym (energy_scale=2.0; pol_modes=coulomb,covariant; tol_abs=1e-10; tol_rel=1e-08
)
Loading model from /Users/kymani/Documents/Big_YM_Data_Sets/Kymani_Colour_Models/5POINT_YM_Model_500
k_Colour/best_model.pt

Running decode mode: beam
  method      : beam
  max_length  : 2153
  beam_size   : 20
  wrote human CSV to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_te
sting/outputs/ym_5pt_colour_500k_beam20_beam_human.csv

[beam] n=1
  top-1 correct   :    0 / 1  (0.0%)
  any-beam correct:    0 / 1  (0.0%)
  avg valid beams : 12.0

Example predictions (1 rows):
  row 0 [beam]
    target simple : (e_1 · e_3*e_2 · p_4*e_4 · e_5)/(2*p_4 · p_5) + (e_1 · e_3*e_2 · e_5*e_4 · p_5)/
(2*p_4 · p_5) - (e_1 · e_3*e_2 · e_4*e_5 · p_4)/(2*p_4 · p_5) - (e_1 · e_3*e_2 · e_4*e_5 · p_5)/(2*p
    input scram   : (e_1 · e_3*e_2 · p_4*e_4 · e_5)/(2*p_4 · p_5) + (e_1 · e_3*e_2 · e_5*e_4 · p_5)/
(2*p_4 · p_5) - (e_1 · e_3*e_2 · e_4*e_5 · p_4)/(2*p_4 · p_5) - (e_1 · e_3*e_2 · e_4*e_5 · p_5)/(2*p
    top1 pred      : -(((p_2·F_3·F_1·p_3·p_4·F_5·F_2·F_4·p_2)/(((p_1·p_2))^2*p_2·p_3·p_2·p_4·p_4·p_5)
))
    top1 metrics  : token=0 string=0 num_simple=0 num_scr=0
    any-beam      : token=0 string=0 num_simple=0 num_scr=0
  wrote detail results to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/da
ta_testing/outputs/ym_5pt_colour_500k_beam20_beam_results.csv

Wrote summary to /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testin
g/outputs/ym_5pt_colour_500k_beam20_summary.csv
Total wall time: 2.98s

Terminal output PDF: /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_te
sting/outputs/Yang-mills_5_Evaluation_Information_2026-09-21_00-42-50-700283.pdf

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